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[1]: temps = [24.4, 25.9, 28.6, 19.3, 26.0, 22.1]
raw = ["24.4", "25.9", "28.6", "19.3"]

temps_f = [t * 9/5 + 32 for t in temps]
print(temps_f)

hot = [t for t in temps if t >= 26]
cool_rounded = [round(t) for t in temps if t < 24]
print(hot, cool_rounded)

floats = [float(s) for s in raw]
print(sum(floats) / len(floats))

[75.92, 78.62, 83.48, 66.74000000000001, 78.8, 71.78]
[28.6, 26.0] [19, 22]
24.55
```

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[2]: results = [("Aline", 85), ("Eric", 78), ("Diane", 91), ("Samuel", 92), ("Grace", 76)]
ranked = sorted(results, key=lambda r: r[1], reverse=True)
print(ranked[:3])

rain_log = [(3, 0.0), (7, 12.5), (11, 3.2), (14, 26.2), (18, 8.9)]
print(max(rain_log, key=lambda d: d[1]))

[('Samuel', 92), ('Diane', 91), ('Aline', 85)]
(14, 26.2)
```

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[4]: import numpy as np
import pandas as pd

df = pd.read_excel("week5_kigali_weather.xlsx")

dates = pd.to_datetime(df["date"], unit="D", origin="1899-12-30").dt.strftime("%Y-%m-%d").to_numpy()
temps = df["temp_c"].to_numpy()
rain = df["rainfall_mm"].to_numpy()
hum = df["humidity_pct"].to_numpy()

print(type(temps), len(temps), dates[0], dates[-1])

<class 'numpy.ndarray'> 60 2026-09-01 2026-10-30
```

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[5]: import numpy as np

print(f"Mean: {temps.mean():.2f} °C")
print(f"Max: {temps.max():.1f} °C")
print(f"Min: {temps.min():.1f} °C")
print(f"Std: {temps.std():.2f} °C")

pos = np.argmax(temps)
print(f"Hottest day: {dates[pos]} at {temps[pos]} °C")

hot_mask = temps >= 26
print(f"Hot days: {hot_mask.sum()}")
print(f"Avg temp on hot days: {temps[hot_mask].mean():.2f} °C")

rainy_mask = rain > 0
print(f"Rainy days: {rainy_mask.sum()}")
print(f"Total rainfall: {rain.sum():.1f} mm")
print(f"Heaviest day: {rain.max():.1f} mm")

print(f"Avg temp, rainy days: {temps[rain > 0].mean():.2f} °C")
print(f"Avg temp, dry days: {temps[rain == 0].mean():.2f} °C")
print(f"Hot AND rainy days: {((temps >= 26) & (rain > 0)).sum()}")
print(f"Days with humidity >= 70: {(hum >= 70).sum()}")

anomaly = temps - temps.mean()
print(f"Days > 3 °C above mean:", dates[anomaly > 3])
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print(f"Days with humidity >= 70: {(hum >= 70).sum()}")

anomaly = temps - temps.mean()
print("Days > 3 °C above mean:", dates[anomaly > 3])
print(f"Days > 3 °C below mean: {(anomaly < -3).sum()}")

first, last = temps[:30].mean(), temps[30:].mean()
print(f"First 30 days: {first:.2f} °C | Last 30 days: {last:.2f} °C")

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Mean: 24.00 °C
Max: 28.6 °C
Min: 19.3 °C
Std: 2.61 °C
Hottest day: 2026-09-17 at 28.6 °C
Hot days: 18
Avg temp on hot days: 27.11 °C
Rainy days: 16
Total rainfall: 331.2 mm
Heaviest day: 26.2 mm
Avg temp, rainy days: 24.14 °C
Avg temp, dry days: 23.95 °C
Hot AND rainy days: 5
Days with humidity >= 70: 16
Days > 3 °C above mean: ['2026-09-07' '2026-09-09' '2026-09-11' '2026-09-12' '2026-09-14'
'2026-09-16' '2026-09-17' '2026-09-19' '2026-09-22']
Days > 3 °C below mean: 8
First 30 days: 26.21 °C | Last 30 days: 21.78 °C

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[6]: wettest = sorted(zip(dates, rain), key=lambda p: p[1], reverse=True)[:5]
for d, r in wettest:
    print(f"{d}: {r} mm")

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2026-09-07: 26.2 mm
2026-09-14: 25.1 mm
2026-09-21: 24.0 mm
2026-09-05: 22.9 mm
2026-09-28: 22.9 mm

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[ ]:

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